University
Name

Global Import-Export Trade Analysis

Trends, Tariffs, and International Trade Relations (1990–2023) — A data analytics project on global trade patterns.

- **PRESENTED BY**
  Student Name
- **GUIDE NAME**
  Prof. Guide Name

1990–
2023



Today's Agenda

Outline of presentation flow for Global Import-Export Trade Analysis (1990–2023)

Introduction to Global Trade

Problem Statement & Objectives

Dataset Overview & Cleaning

Methodology & Tools

Visual Analytics

Import & Export Trends

Tariffs & Product Analysis

Country Growth Comparisons

World Trade Relations Map

Findings, Recommendations & Q&A



1990–
2023

GLOBAL TRADE
CONTEXT

Introduction: Global Trade

WHY GLOBAL TRADE ANALYSIS MATTERS

International trade has been a cornerstone of economic development for decades. From 1990 to 2023, global trade volumes expanded dramatically, reshaping economies, geopolitical alliances, and industrial strategies worldwide.

Global trade drives economic growth and development
Influences international relations and policy decisions
Trade patterns guide business and government strategy
Globalization accelerated cross-border commerce post-1990

Problem Statement

CHALLENGES & GAPS ADDRESSED IN GLOBAL TRADE ANALYSIS

- Complexity of global trade data requires robust analytics to extract meaningful insights across 30+ years of records.
- Tariff policies and trade relations are constantly evolving, making static analysis insufficient for policy decisions.
- Lack of comprehensive, data-driven frameworks to simultaneously assess import-export trends, tariff structures, and comparative advantages.
- Existing analyses often lack actionable recommendations for both policymakers and businesses navigating international trade.



Project Objectives

KEY RESEARCH GOALS

TREND ANALYSIS

Analyze import and export value trends over the period 1990–2023.

TOP TRADING NATIONS

Identify top trading countries and their product share in global trade.

TARIFF EXAMINATION

Examine tariff structures, trade balances, and policy implications.

TREND ANALYSIS

01 Track the evolution of global import and export values from 1990 to 2023, identifying periods of growth, slowdown, and recovery linked to major global economic events and policy shifts.

GROWTH & RCA ASSESSMENT

02 Assess each country's trade growth relative to global trends and evaluate Revealed Comparative Advantage (RCA) to highlight sectors where countries hold strong international competitiveness.

RECOMMENDATIONS

03 Provide actionable, data-driven recommendations for policymakers and businesses to support trade facilitation, export diversification, and sustainable economic growth strategies.



Dataset Overview

OVERVIEW OF DATASETS, SYSTEMS, AND INFORMATION USED ↗

Source: World Bank — official global trade datasets covering imports, exports, and tariff data across 180+ countries.
Time Period: 1990 to 2023 (33 years of longitudinal trade data)
Key Variables: Import Value (USD), Export Value (USD), MFN Tariff Rate, Applied Tariff Rate, Product Categories (HS Codes), Country Codes, Year
Scope: Multi-country, multi-year panel data enabling trend analysis, tariff comparison, and trade balance computation across all major economies and product sectors.

50K+

Trade records from World Bank datasets

Data Cleaning Process

Rigorous data preparation was applied to World Bank trade datasets (1990–2023) to ensure reliability, consistency, and analytical readiness across all import, export, and tariff variables.

01



HANDLE MISSING VALUES

Applied imputation for estimable gaps and removed records with critical missing trade values to ensure dataset completeness and integrity.

02



REMOVE DUPLICATE RECORDS

Identified and eliminated duplicate entries across country-year records to maintain accuracy and prevent skewed analytical results.

03



CONVERT DATA TYPES

Standardized numeric, date, and categorical fields for consistency, enabling seamless computation and cross-variable comparisons.

04

ENGINEER NEW FEATURES

Derived trade balance, year-on-year growth rates, and RCA indicators to enrich the dataset for deeper analytical insights.

Methodology

TOOLS AND TECHNIQUES USED IN THE PROJECT

01

PYTHON & PANDAS

Python with Pandas & NumPy for data manipulation and processing

02

PLOTLY VISUALIZATION

Plotly for interactive, high-quality visual analytics and dashboards

03

STATISTICAL METHODS

Correlation analysis and RCA calculation for trade insights

04

DATA STORYTELLING

Translating analytics into actionable business and policy insights

Import Value Trend

GLOBAL IMPORT VALUE ANALYSIS OVER TIME (1990–2023)

- Overall upward trend in global import values from 1990 to 2023, driven by trade liberalization.
- Significant growth acceleration post-2000 with globalization and emerging market integration.
- Notable slowdowns during 2008 global financial crisis and 2020 COVID-19 pandemic disruptions.
- Import trends reflect economic health, policy impact, and shifting global demand patterns.



Export Value Trend

OBJECTIVE: ANALYZE EXPORT GROWTH PATTERNS AND VOLATILITY (1990–2023)

- Exports generally mirror imports with distinct country-level variances across regions.
- Emerging markets show stronger export growth since 2000, outpacing developed economies.
- Export sectors respond dynamically to global demand shifts and supply chain changes.
- Export performance is critical for trade surplus, GDP growth, and economic development.



Import vs Export

ASSESSING TRADE BALANCES, SURPLUSES & DEFICITS BY COUNTRY ↗

Trade surpluses and deficits vary significantly by country and time period. Persistent trade deficits may signal underlying competitiveness issues, while surpluses reflect strong export capacity.

- USA consistently among largest import-deficit nations
- China and Germany maintain strong export surpluses
- Emerging markets show growing export competitiveness since 2000
- Trade imbalances directly influence currency valuations and monetary policy
- Bilateral trade gaps drive diplomatic and tariff negotiations



Tariff Distribution

MFN VS APPLIED TARIFFS: GLOBAL POLICY COMPARISON

- MFN Tariffs: Standard rates applied to all WTO members equally
- Applied Tariffs: Actual rates after trade agreements & preferences
- Developed countries maintain lower applied tariffs (~3–5%)
- Developing countries show higher applied tariffs (~10–15%)
- Tariff reductions through FTAs drive trade volume growth
- Disparities in tariff structures reflect policy complexity and negotiation leverage
- Lower tariffs correlate with higher trade competitiveness and market access



~5%

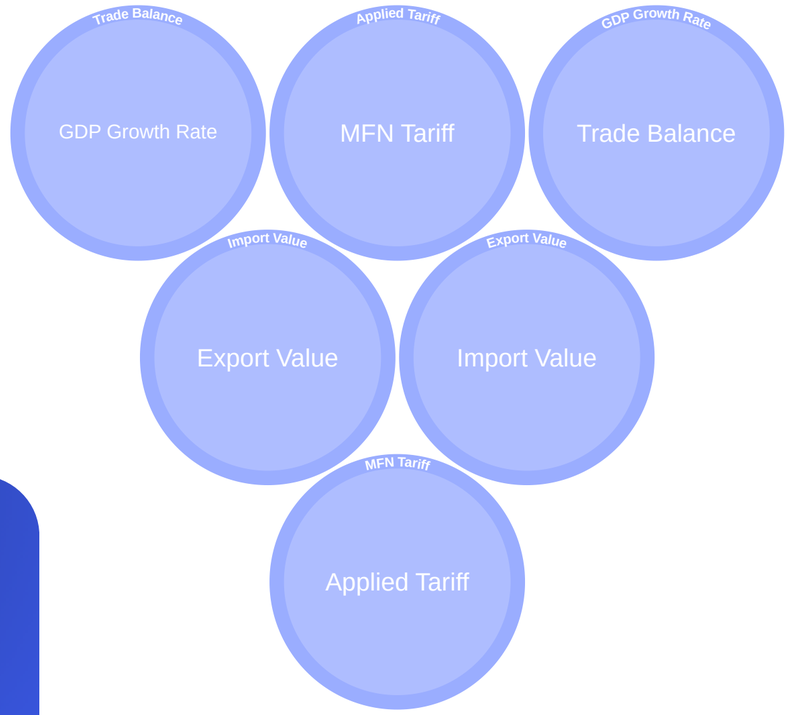
AVG APPLIED TARIFF IN DEVELOPED NATIONS

Correlation Heatmap

STATISTICAL RELATIONSHIPS BETWEEN TARIFFS, TRADE VALUES, GROWTH & BALANCE

Key Insights:

- Tariffs are inversely correlated with trade volumes — lower tariffs drive higher trade flows
- Export growth shows strong positive correlation with GDP growth
- Trade balance correlates negatively with import dependency
- Applied tariffs and MFN tariffs exhibit high mutual correlation
- Correlations guide critical policy leverage points for trade facilitation



-0.72

TARIFF VS TRADE VOLUME CORRELATION

01

Trade Growth Trend

Global trade grew substantially 1990–2023 with notable volatility during global crises.

02

Country Dominance

Top economies lead imports and exports; emerging markets steadily gaining global share.

03

Tariffs & RCA

Tariff policies vary widely; RCA reveals competitive sectors and strong trade partnerships.

04

Recommendations

Pursue tariff harmonization, export diversification, and data-driven market strategies.

Major Findings & Recommendations

KEY INSIGHTS, ACTIONABLE RECOMMENDATIONS & CONCLUSION

Global trade grew substantially from 1990 to 2023, with volatility during crises such as 2008 and 2020. Top economies dominate imports and exports, while emerging markets are gaining significant share. Tariff policies vary widely, influencing trade costs and competitiveness. RCA and trade relations highlight competitive sectors and strong bilateral partnerships. Governments should pursue tariff harmonization and trade facilitation measures. Policymakers must support export diversification based on RCA insights. Businesses should leverage data analytics for targeted market entry and supply chain resilience. Data analytics provides critical insights into complex global trade dynamics, enabling evidence-based policy and business decisions to foster sustainable and inclusive growth.

Thank You!

Questions & Answers



Trade
Analytics

Global Import-Export Trade Analysis in USD: Trends, Tariffs, and International Trade Relations (1990–2023)
Data Analytics Project | University Seminar

1990
– 2023